

Generation Interconnection Request System Impact Study Report

For Queue Project Q11-0001

N. Shannon 138 kV
September 19, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0001. The Interconnection Customer proposes a 204.3 MW wind facility interconnecting at the N. Shannon 138 kV substation in the Kentucky area. The request was received on January 01, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 0 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$275,594,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Toledo Bend – Killen 500kV, the Killen 500-230 kV Autotransformer loads to 134.4% of its 796 MVA rating. Project DFAX: 7.9%.

Toledo Bend – Wharton 230kV Line: 117.0% loading (522 MVA) following the outage of the Toledo Bend – Adkins 500kV; DFAX = 11.4%.

For the loss of Dekalb – Jones Chapel 500kV L, the Russell Fork – Robison Park 230kV Line loads to 117.0% of its 557 MVA rating. Project DFAX: 7.6%.

Alibates – Action 138kV: 116.1% loading (382 MVA) following the outage of the Pine Bluff Watson Chapel – Carraizo 1201 345kV; DFAX = 25.5%.

For the loss of Dekalb – Jones Chapel 500kV line, the Concord – Bluebonnet Substation 230kV loads to 115.9% of its 499 MVA rating. Project DFAX: 5.2%.

For the loss of Russell Fork – Robison Park 230kV, the Bluebonnet – Jones Chapel 230kV Line loads to 114.6% of its 575 MVA rating. Project DFAX: 5.1%.

The Killen – Moffat 230kV is overloaded to 113.7% of its 773 MVA emergency rating for the loss of the Killen Substation – Moffat 500kV line. The project contributes 22.0% DFAX (45.0 MW) to this overload.

For the loss of Harvest Power Orlando – Frontier 138kV, the Sptsum Lornex – Latah Jct Substation 138kV line loads to 113.3% of its 257 MVA rating. Project DFAX: 19.5%.

For the loss of McClellan Ar – Spatsum Lornex 138kV, the Yuba City Sta A – Westin-Elgin 138kV loads to 110.4% of its 329 MVA rating. Project DFAX: 14.5%.

The McClellan Ar Substation – Westin-Elgin Substation 138kV is overloaded to 109.0% of its 286 MVA emergency rating for the loss of the Yuba City Sta A – Westin-Elgin 138kV. The project contributes 11.6% DFAX (23.8 MW) to this overload.

For the loss of Toledo Bend – Killen 500kV L, the Ironman Substation – Killen 230kV loads to 107.9% of its 779 MVA rating. Project DFAX: 14.1%.

For the loss of Pine Bluff Watson Chapel – Carraizo 1201 345kV L, the Eldorado – Action 138kV loads to 107.1% of its 379 MVA rating. Project DFAX: 25.5%.

For the loss of Westin-Elgin – McClellan Ar 138kV, the Temple Sta 243 – McClellan Ar Substation 138kV loads to 106.6% of its 217 MVA rating. Project DFAX: 9.8%.

For the loss of Jones Chapel Substation – Dekalb 500kV, the Russell Fork – Bluebonnet 230kV loads to 105.7% of its 486 MVA rating. Project DFAX: 5.5%.

For the loss of Cooper – Universal Studios Tap Substation 138kV, the Brunswick Cellulose – Shannon Farm 138kV loads to 105.4% of its 265 MVA rating. Project DFAX: 10.3%.

The McClellan Ar – Spatsum Lornex 138kV is overloaded to 104.5% of its 366 MVA emergency rating for the loss of the Yuba City Sta A – Westin-Elgin 138kV line. The project contributes 15.8% DFAX (32.4 MW) to this overload.

For the loss of Latah Jct – Spatsum Lornex 138kV L, the Frontier – Harvest Power Orlando 138kV loads to 104.4% of its 253 MVA rating. Project DFAX: 16.2%.

The Alibtes – Latah Jct 138kV is overloaded to 104.0% of its 270 MVA emergency rating for the loss of the Harvest Power Orlando – Frontier 138kV. The project contributes 19.5% DFAX (39.7 MW) to this overload.

Temple Station 243 – Westin-Elgin 138kV line: 103.8% loading (251 MVA) following the outage of the Westin-Elgin – McClellan Ar 138kV; DFAX = 9.8%.

The Yuba City Sta A – Frontier 138kV is overloaded to 103.6% of its 340 MVA emergency rating for the loss of the McClellan Ar – Spatsum Lornex 138kV L. The project contributes 14.5% DFAX (29.6 MW) to this overload.

Brunswick Cellulose Substation – Bozeman S Substation 138kV: 101.1% loading (260 MVA) following the outage of the Universal Studios Tap – Cooper Substation 138kV Line; DFAX = 10.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Killen 500/230 kV Autotransformer	Replace transformer with higher-rated unit	\$17,719,000	100% (\$17,719,000)
Toledo Bend – Wharton 230kV	Rebuild line with higher-capacity conductor	\$10,095,000	100% (\$10,095,000)
Russell Fork – Robison Park 230kV Line	Rebuild line with higher-capacity conductor	\$52,936,000	100% (\$52,936,000)
Action – Alibtes 138kV	Rebuild line with higher-capacity conductor	\$25,329,000	100% (\$25,329,000)
Concord – Bluebonnet Substation 230kV	Rebuild line with higher-capacity conductor	\$17,090,000	100% (\$17,090,000)
Bluebonnet – Jones Chapel 230kV line	Rebuild line with higher-capacity conductor	\$37,035,000	100% (\$37,035,000)
Killen – Moffat 230kV	Rebuild line with higher-capacity conductor	\$16,674,000	100% (\$16,674,000)
Spatsum Lornex – Latah Jct 138kV line	Rebuild line with higher-capacity conductor	\$7,903,000	100% (\$7,903,000)
Yuba City Sta A – Westin-Elgin 138kV	Rebuild line with higher-capacity conductor	\$25,795,000	100% (\$25,795,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mcclellan Ar – Westin-Elgin 138kV	Reconductor line	\$3,179,000	100% (\$3,179,000)
Ironman – Killen Substation 230kV	Reconductor line	\$8,255,000	100% (\$8,255,000)
Acion – Eldorado 138kV	Reconductor line	\$1,229,000	100% (\$1,229,000)
Temple Sta 243 – Mcclellan Ar 138kV	Reconductor line	\$2,315,000	100% (\$2,315,000)
Bluebonnet – Rssell Fork 230kV line	Reconductor line	\$20,520,000	100% (\$20,520,000)
Brunswick Cellulose – Shannon Farm 138kV L	Reconductor line	\$1,595,000	100% (\$1,595,000)
Mcclellan Ar – Spatsum Lornex 138kV	Reconductor line	\$4,540,000	100% (\$4,540,000)
Harvest Power Orlando Substation – Frontier 138kV	Reconductor line	\$8,707,000	100% (\$8,707,000)
Alibtes – Ltah Jct 138kV	Reconductor line	\$3,598,000	100% (\$3,598,000)
Temple Station 243 – Westi-Elgin 138kV line	Reconductor line	\$1,512,000	100% (\$1,512,000)
Yuba City Sta A – Frontier 138kV	Reconductor line	\$7,006,000	100% (\$7,006,000)
Brunswick Cellulose Substation – Bozeman S Substation 138kV	Reconductor line	\$2,562,000	100% (\$2,562,000)
	Total allocated to project		\$275,594,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.